

15 marks

Pg 18

// Intersection between two given set

import java.util.*;

public class p18 {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

Set<Integer> set1=new HashSet<>();

Set<Integer> set2=new HashSet<>();

for(int i=0;i<4;i++)

{

set1.add(sc.nextInt());

}

System.out.println("\n");

for(int i=0;i<5;i++)

{

set2.add(sc.nextInt());

}

```

        set1.retainAll(set2);

        System.out.println("The intersection of the sets: set1 and
set2"+" ":"+set1);
    }
}

```

P19

/*Find the Largest Word in a Sentence

Write a Java program that takes a sentence as input and determines the largest word in that sentence based on its length.

If there are multiple words with the same maximum length, the program should return the first one encountered.

```

*/
import java.util.*;

class p19
{
    public static void main(String args[])

    {
        Scanner sc=new Scanner(System.in);
        String s=sc.nextLine();
        String[] arr=s.split(" ");
        String large="";

```

```

    for(int i=0;i<arr.length;i++)
    {
        if(arr[i].length()>large.length())
        {
            large=arr[i];
        }
    }

    System.out.println("The largest string word is: "+" "+large);

}
}

```

Pg20

/*Extract the Last Character of Each Word in a Sentence

Write a Java program that takes a sentence as input and extracts the last character of each word in that sentence.

The program should concatenate these last characters into a single string and display the result. */

```

import java.util.*;

class p20
{
    public static void main(String args[])

```

```

{
    Scanner sc=new Scanner(System.in);
    String s=sc.nextLine();
    String[] arr=s.split(" ");
    String last="";

    for(int i=0;i<arr.length;i++)
    {
        if(!arr[i].isEmpty())
        {
            last=last+arr[i].charAt(arr[i].length()-1);

        }

    }

    System.out.println("The list of last words are:"+ " "+last);

}
}

```

// Majority of an element means if an element is present in an array for more than $n/2$ times,

// where n is the length of the Array

// Array = {3, 3, 4, 2, 4, 4, 2, 4, 4} => 4

import java.util.*;/**

*** p21**

***/**

class p21 {

public static void main(String args[])

{

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int arr[]=new int[n];

for(int i=0;i<arr.length;i++)

{

arr[i]=sc.nextInt();

sc.nextLine();

}

int can=arr[0];

int count=0;

for(int i=0;i<n;i++)

```
{  
    if(count==0)  
    {  
        can=arr[i];  
    }  
  
    if(arr[i]==can)  
    {  
        count=count+1;  
    }  
    else  
    {  
        count--;  
    }  
}
```

```
count=0;  
for(int i=0;i<n;i++)  
{  
    if(arr[i]==can)  
    {  
        count++;  
    }  
}
```

```

    if(count>n/2)
    {
        System.out.println("the majority element is: "+can);
    }

    else {
        System.out.println("There is no majority element.");
    }
}

```

Pg21

// Majority of an element means if an element is present in an array for more than $n/2$ times,

// where n is the length of the Array

// Array = {3, 3, 4, 2, 4, 4, 2, 4, 4} => 4

import java.util.*;/**

*** p21**

***/**

class p21 {

```
public static void main(String args[])
{
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int arr[]=new int[n];
    for(int i=0;i<arr.length;i++)
    {
        arr[i]=sc.nextInt();
        sc.nextLine();
    }
```

```
int can=arr[0];
int count=0;
for(int i=0;i<n;i++)
{
    if(count==0)
    {
        can=arr[i];
    }
```

```
    if(arr[i]==can)
    {
        count=count+1;
```



```
    }  
    else  
    {  
        count--;  
    }  
}
```

```
count=0;  
for(int i=0;i<n;i++)  
{  
    if(arr[i]==can)  
    {  
        count++;  
    }  
}
```

```
if(count>n/2)  
{  
    System.out.println("the majority element is: "+can);  
}
```

```
else {  
    System.out.println("There is no majority element.");  
}  
}
```

/* Marge two strings using comma

Input:

Arijit,Ghosh

Pop:Mom

Output:

Arijit, Ghosh, Pop:Mom

***/**

import java.util.*;

class p22

{

public static void main(String args[])

{

Scanner sc=new Scanner(System.in);

String s1=sc.nextLine();

String s2=sc.nextLine();

String result=String.join(", ",s1,s2);

System.err.println(result);

}

}

}

Pg23

/* Find the Maximum Ascii valued character in a String

*** Arijit = t**

*** priyanka = y**

*** Aa = a**

***/**

import java.util.*;

class p23

{

public static void main(String args[])

{

Scanner sc= new Scanner(System.in);

String str=sc.nextLine();

//Char ch=sc.nextCharacter();

char ch=str.charAt(0);

for(int i=0;i<str.length();i++)

{

if(ch<str.charAt(i))

{

ch=str.charAt(i);

```
    }  
}
```

```
    System.out.println("character with highest ascii value:"+ch);
```

```
}
```

```
}
```

Pg26

/* Find the occurrence of each character in a string

*** Input: Hello world**

*** Output:**

*** H: 1**

e: 1

l: 3

o: 2

w: 1

r: 1

d: 1

***/**

```
import java.util.*;

class pg26
{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String str1=sc.nextLine().replace(" ", "");
        LinkedHashMap<Character,Integer> count= new LinkedHashMap<>();

        for(int i=0;i<str1.length();i++)
        {
            if(!count.containsKey(str1.charAt(i)))
            {
                count.put(str1.charAt(i),1);
            }

            else{
                count.put(str1.charAt(i),count.get(str1.charAt(i))+1);
            }
        }

        for(int i=0;i<count.size();i++)
        {
            Character key= (Character)count.keySet().toArray()[i];
            Integer value=count.get(key);
```

```
System.out.println(key+":"+value);
```

```
}
```

```
}
```

```
}
```

Pg27

```
/* Take a number. If it odd then reverse it, otherwise print "Can not reverse"
```

```
*
```

```
* Input : 123
```

```
* Output : 321
```

```
*
```

```
* Input : 234
```

```
* Output : Can not reverse
```

```
*/
```

```
import java.util.*;
```

```
class p27
```

```
{
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int a=sc.nextInt();
```

```
        int sum=0;
```

```
int temp;
if(isOdd(a)==true)
{
    while(a!=0)
    {
        temp=a%10;
        sum=sum*10+temp;
        a=a/10;
    }

    System.out.println(sum);

}

else
{
    System.out.println("Can not reverse");
}

}

public static boolean isOdd(int num)
{
    if(num%2!=0)
    {
```

```
        return true;
    }

    else{
        return false;
    }
}

}
```

Pg30

/* Remove Vowel From A String

*** Input : Arijit Ghosh**

*** Output : rjt Ghsh**

***/**

```
import java.util.*;

class p30
{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String str1=sc.nextLine().toLowerCase();
        char arr[]={'a','e','i','o','u'};
        String str2="";
```



```

for(int i=0;i<str1.length();i++)
{
    int j;
    for(j=0;j<arr.length;j++){
        if(str1.charAt(i) == arr[j])
        {
            break;

        }
    }
    if(j==arr.length){
        str2=str2+str1.charAt(i);
    }
}

```

```

System.out.println("string is: "+" "+ str2);

```

```

}
}

```

Pg 31

/*

* You are given a string S consisting of lowercase letters. Your task is to remove all the consecutive duplicates from

* the string and output the result. For example, if the input string is "abbcddfeff", the output should be "abcdef".

*** Input: abbcdeff**

*** Output: abcdef**

***/**

import java.util.*;

class pg31

{

public static void main(String[] args) {

Scanner sc=new Scanner(System.in);

String str1=sc.nextLine();

String result=getConsecutiveDuplicates(str1);

System.out.println(result);

}

public static String getConsecutiveDuplicates(String s)

{

String str2="";

char ch='\0';

for(int i=0;i<s.length();i++)

{

if(s.charAt(i)!=ch)

{

ch = s.charAt(i);

str2+= ch;

}

```
}  
return str2;  
}
```

```
}
```

```
/*import java.util.Scanner;
```

```
class RemoveConsecutiveDuplicates {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a string: ");  
        String input = sc.nextLine();  
  
        // Initialize an empty string to store the result  
        String result = "";  
  
        // Loop through the input string  
        for (int i = 0; i < input.length(); i++) {  
            // Check if the current character is the first character or not the  
            same as the last character in the result  
            if (i == 0 || input.charAt(i) != input.charAt(i - 1)) {  
                result += input.charAt(i); // Append the character to the result  
            }  
        }  
    }  
}
```

```

    }

    System.out.println("Output: " + result); // Print the result
}
}
*/

```

Pg32

```

/* Remove a character from a string
* Input
* -----
* Arijit
* i
* Output
* -----
* Arjt
*/
import java.util.*;
class p32
{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);

        String str1=sc.nextLine();
        String ch=sc.nextLine();
    }
}

```

```
String s=str1.replace(ch,"");
```

```
System.out.println(s);
```

```
}
```

```
}
```

Pg33

/*Write a program that takes a string as input where letters are followed by numbers, and prints the expanded version of the string.

Each letter should be repeated according to the number that follows it.

For example, if the input is a2b3c1, the output should be aabbbc. */

```
/*import java.util.*;
```

```
class p33
```

```
{
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        String s=sc.nextLine();
```

```
        for(int i=0;i<s.length();i++)
```

```
        {
```

```
            if(Character.isAlphabetic(s.charAt(i)))
```

```
            {
```

```
                System.out.print(s.charAt(i));
```

```
            }
```

```

        else if(Character.isDigit(s.charAt(i)))
        {
            int x=Integer.parseInt(String.valueOf(s.charAt(i)));
            for(int j=0;j<x;j++)
            {
                System.out.print(s.charAt(i));
            }
        }
    }
}

}

*/

//if u are to repeat string repeats the entire string, not just the letter

import java.util.*;

class p33
{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String s=sc.nextLine();
        for(int i=0;i<s.length();i++)
        {

```

```

        if((Character.isAlphabetic(s.charAt(i))))
        {
            System.out.println(s.charAt(i));
        }

        else if(Character.isDigit(s.charAt(i)))
        {
            int r=Integer.parseInt(String.valueOf(s.charAt(i)));
            String res=s.repeat(r);
            System.out.println(res);
        }

    }

}
}

```

Pg34

/* qn 34 Calculate the sum of all digits of a numbers. If the sum is greater than 9 then again sum the digits. The sum is continuing until the sum becomes one digit.

Input : 195

Output : 6

Explanation: $1+9+5 = 15 > 9$, so $1+5=6$ */

```
import java.util.*;

class p34
{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt();
        int sum=0;

        if(a<=9)
        {
            System.out.println(a);
        }
        else{
            int res=sumdigi(a);
            System.out.println(res);
        }
    }

    public static int sumdigi(int a)
    {
        int div=a;
        int sum=0;
        int rev;
        while(div>0)
```



```

    {
        rev=div%10;
        sum=sum+rev;
        div=div/10;
    }

    if(sum>9)
    {
        return sumdigi(sum);
    }

    else
    {
        return sum;
    }
}

```

Pg35

/* Input: "Hello@123World456"

qn 35

Process:

Array: ["Hello", "@", "123", "World", "456"]

Now you should concatenate all the strings in the array and multiply it with the sum of the position of that array where numbers are stored.

////////////////////////////////////

```

String con="";
for(int i=0;i<str.length;i++)
{
    con=con+str[i];
}

int sum=0;
///-----//
for(int i=0;i<con.length();i++)
{
    if(Character.isDigit(con.charAt(i)))
    {
        sum=sum+(i+1);
    }
}

//-----multiply the result-----//
String result=con.repeat(sum);
System.out.println("The result is:"+ " "+result);

}
}

```

Pg36

```
/* Reverse of a string qn 36
* Arijit = tijirA
*/
import java.util.*;
class p36
{
    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        String s=sc.nextLine().toLowerCase();
        for(int i=s.length()-1;i>=0;i--)
        {
            System.out.print(s.charAt(i));

        }
    }
}
```

Pg37

/*Count the number of words in a sentence that start and end with the same letter. qn 37

Input: Anna asked about the Ginseng recipe

Output: 2

Explanation: There are two words in the sentence that start and end with the same letter - "Anna" and "Ginseng". */

```
import java.util.*;
```

```
class p37
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        String s=sc.nextLine().toLowerCase();
```

```
        String str[]=s.split(" ");
```

```
        int count=0;
```

```
        for(int i=0;i<str.length;i++)
```

```
        {
```

```
            if(str[i].charAt(0)==str[i].charAt(str[i].length()-1))
```

```
            {
```

```
                count=count+1;
```

```
            }
```

```
        }
```

```
        System.out.println("count"+" "+count);
```

```
    }
```

```
}
```

Pg38.java

```
/* Minimum Ascii valued character in a String */
import java.util.*;

class p38
{
    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        String str=sc.nextLine();
        char c=str.charAt(0);
        for(int i=1;i<str.length();i++)
        {
            if(str.isEmpty())
            {
                return;
            }
            if(Character.compare(c,str.charAt(i))>0)
            {
                c=str.charAt(i);
            }
        }
        if (c == ' ') {
            System.out.println("Minimum ASCII valued character: [space]");
        } else {
            System.out.println("Minimum ASCII valued character: " + c);
        }
    }
}
```

```

    }
}
}

//To make sure the program clearly indicates that a space character is the
minimum,

// we can modify the output to explicitly mention if the minimum
character is a space.

/* -----splitting the string with space

import java.util.*;

class p38 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        String str = sc.nextLine();

        // Split the string by spaces and store the parts in an array
        String[] parts = str.split(" "); // Split by space

        // Initialize minimum character with the first non-empty part
        char minChar = Character.MAX_VALUE; // Set to maximum char value
        to ensure any char will be smaller

        // Iterate through each part to find the minimum ASCII valued
        character

```

```

    for (String part : parts) {
        for (int i = 0; i < part.length(); i++) {
            char currentChar = part.charAt(i);
            if (currentChar < minChar) {
                minChar = currentChar; // Update minChar if currentChar is
smaller
            }
        }
    }

    // Check if minChar was updated
    if (minChar == Character.MAX_VALUE) {
        System.out.println("No characters to compare.");
    } else {
        // Determine if the minimum character is a space
        System.out.println("Minimum ASCII valued character: " + minChar);
    }
}

```

or another method:

```
import java.util.Scanner;
```

```
class MinimumAsciiCharacter {
```



```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Enter a string: ");  
    String input = sc.nextLine();  
  
    // Call the method to find the minimum ASCII valued character  
    char minChar = findMinAsciiCharacter(input);  
  
    // Print the result  
    System.out.println("Minimum ASCII valued character: " + minChar);  
}
```

```
private static char findMinAsciiCharacter(String s) {  
    // Initialize the minimum character with the first character of the  
    string  
    char minChar = s.charAt(0);  
  
    // Iterate through the string to find the minimum ASCII character  
    for (int i = 1; i < s.length(); i++) {  
        if (s.charAt(i) < minChar) {  
            minChar = s.charAt(i); // Update minChar if a smaller ASCII  
            character is found  
        }  
    }  
  
    return minChar;
```

```
}  
}
```

```
*/
```

Pg39

/* Sort name by dictionary order

Input

Arijit

Minaz

Arnab

Output

Arijit

Arnab

Minaz

```
*/
```

```
import java.util.*;
```

```
class p39
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        TreeMap<String,Integer> map=new TreeMap<>();
```

```

    for(int i=0;i<3;i++)
    {
        String str=sc.nextLine();
        map.put(str,i);
    }
    System.out.println("-----");

    //convert sorted keys to array
    String[] sorted=map.keySet().toArray(new String[0]);
    //printing
    for(int i=0;i<sorted.length;i++)
    {
        System.out.println(sorted[i]);
    }

}
}

```

Pg40

/* Count number of spaces and characters in a string

*** Input**

*** -----**

*** Arijit is bad boy**

*** Output**

*** -----**

*** Number of space : 3**

Number of character : 14

***/**

import java.util.*;

class p40

{

public static void main(String args[])

{

Scanner sc=new Scanner(System.in);

String str=sc.nextLine();

int count=0;

for(int i=0;i<str.length();i++)

{

if(str.charAt(i)==' ')

{

count=count+1;

}

}

**System.out.println("The no of spaces:"+count+"and"+" "+"no of
charcters:"+(str.length()-count));**

}

}

Pg43

/* Count the sum of even numbers in a given number

***/**

import java.util.*;

class p43

{

public static void main(String args[])

{

Scanner sc=new Scanner(System.in);

String s=sc.nextLine();

int sum=0;

for(int i=0;i<s.length();i++)

{

int b=Integer.parseInt(String.valueOf(s.charAt(i)));

for(i=0;i<s.length()+1;i++){

if(b%2==0)

{

sum=sum+b;

}

}

}

```
        System.out.println("Sum of evn digits:"+" "+sum);
    }
}
```

Pg44

```
/* Count the sum of prime digits in a given number
```

```
*
```

```
* Input: 2345678910
```

```
* Output: 17
```

```
*
```

```
* Explanation: 2+3+5+7 = 17
```

```
*/
```

```
import java.util.*;
```

```
class p44
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        String s=sc.nextLine();
```

```
        int sum=0;
```

```
        for(int i=0;i<s.length();i++)
```

```
        {
```

```
            int b=Integer.parseInt(String.valueOf(s.charAt(i)));
```

```
            for(i=2;i<b+1;i++){
```

```
                if(b%i!=0)
```

```
        {  
            sum=sum+b;  
        }  
  
    }  
  
}  
System.out.println("Sum of prime digits:"+ " "+sum);  
}  
}
```